

Small-Batch Training with AdamW

Synchronous and four-worker
decentralized training

How do batch size, moment memory, and gradient clipping
affect validation quality at a fixed token budget?

01 BATCH & MEMORY

02 DECENTRALIZATION

03 CLIPPING

One token budget, two optimization systems

MODEL, DATA, AND EVALUATION

20.4M parameters

C4 · context 512 · 40 virtual epochs

408,092,672 global training targets

20,447,232 warmup targets

197,411,295 final validation targets

SHARED OPTIMIZER SETTINGS

AdamW · weight decay 0.1 · $\varepsilon = 10^{-8}$

Cosine LR decay to zero

FP32 parameters / moments · BF16 AMP

GLOBAL BATCH: 16, 32, 64, 128

Synchronous

One model, one gradient, one Adam state

Four-worker decentralized

Local batches: 4, 8, 16, 32

Independent moments; reciprocal AWC

Evaluate the FP32 mean of four models

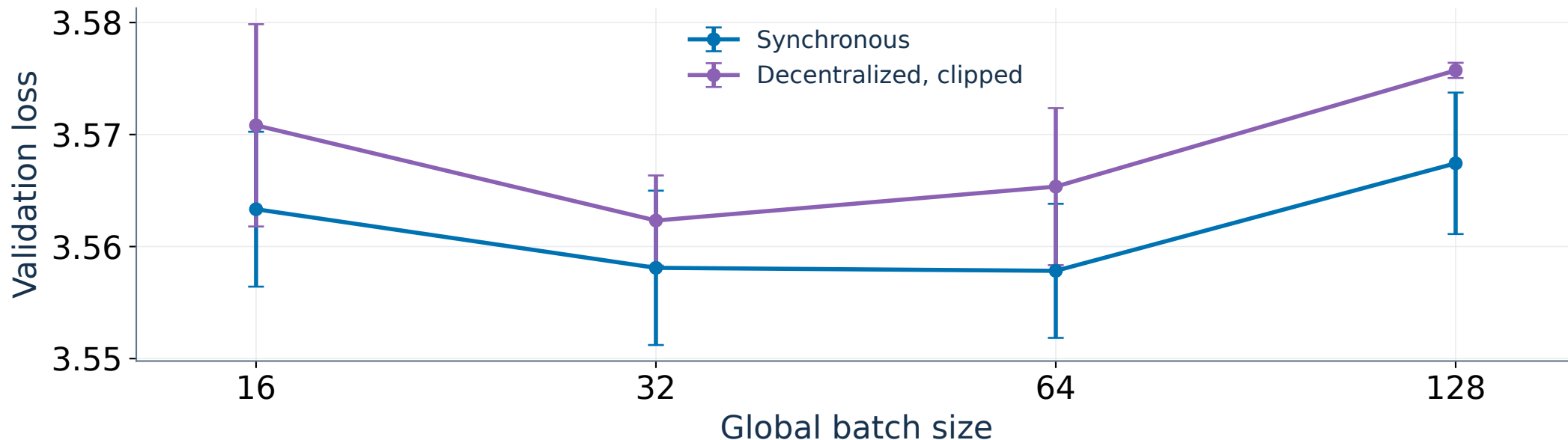
Four logical workers share one GH200.

Same training targets and full validation; different tuning procedures.

Synchronous: conditional LR / β_2 then β_1 tuning. Decentralized: joint grids and frozen replication selections.

No adaptive consensus, checkpoints, or saved weights. All quality comparisons are exploratory.

Small batches can improve validation quality



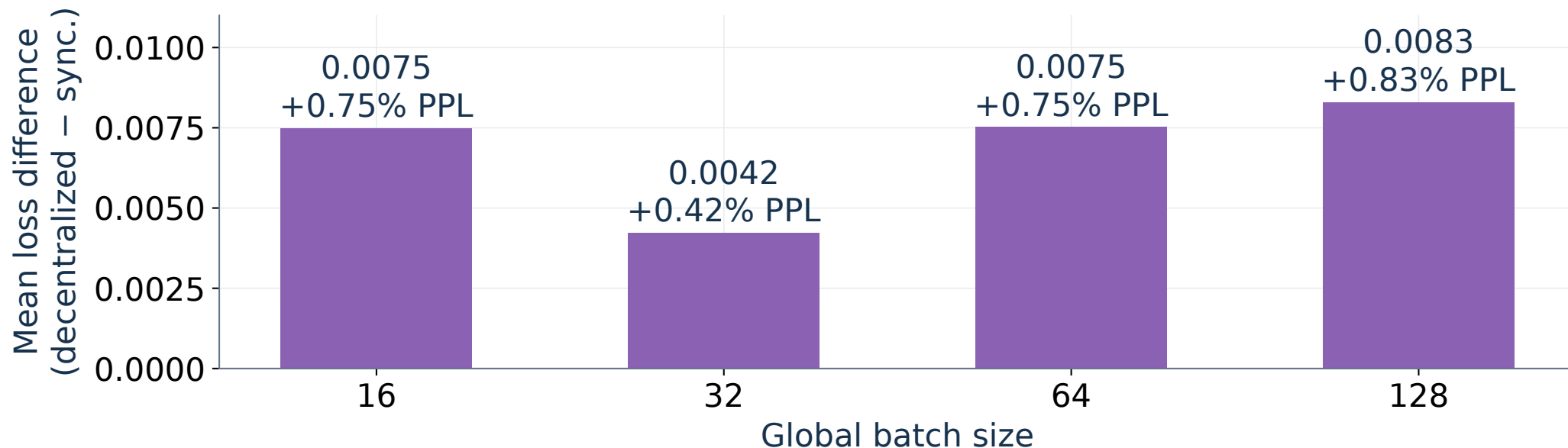
3.557833 synchronous · batch 64

3.562323 decentralized · batch 32

Improvement over batch 128: 0.009597 synchronous; 0.013404 decentralized.

Lower loss is better. Batch 16 is not best. Bars: sample SD over seeds 42–44, not confidence intervals.

Decentralized training approaches synchronous quality

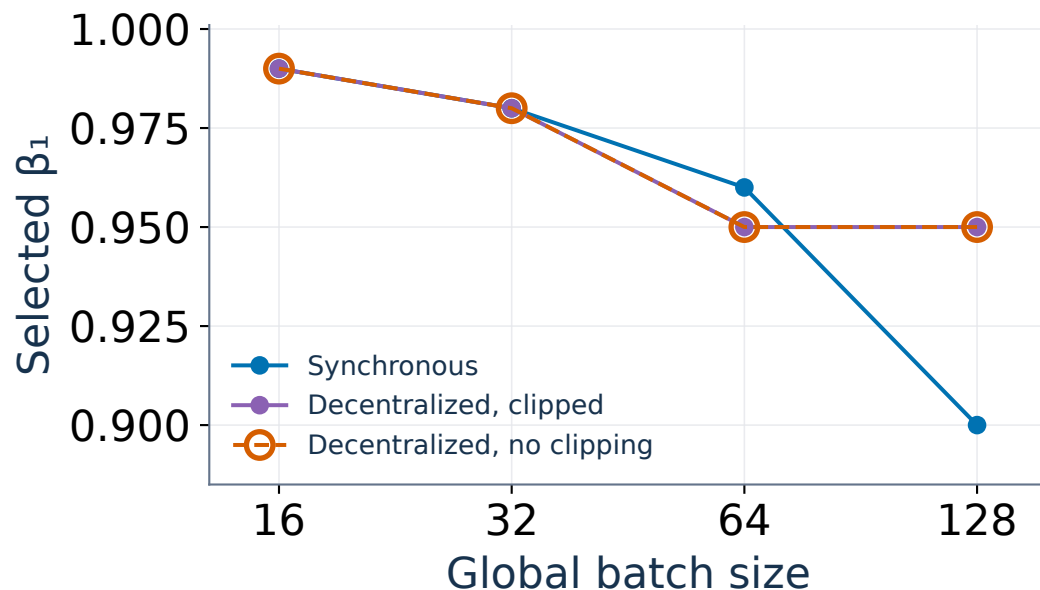


0.004224–0.008296 loss gap

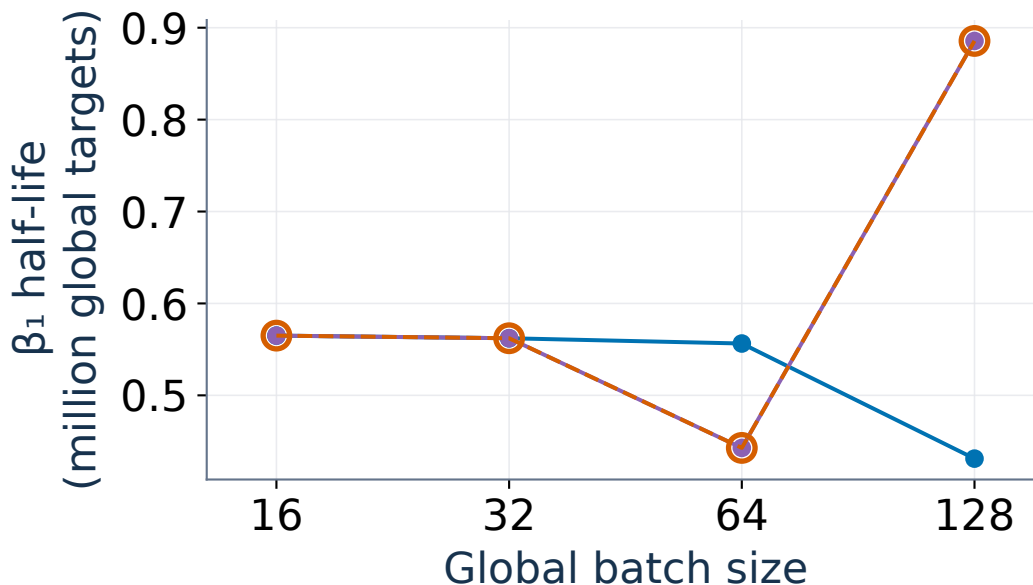
0.42–0.83% higher perplexity

Clipping threshold 1 for both methods. These are observed differences between selected recipes, not a statistical equivalence test. Identical seed numbers do not imply identical trajectories.

Higher β_1 preserves memory as batches shrink



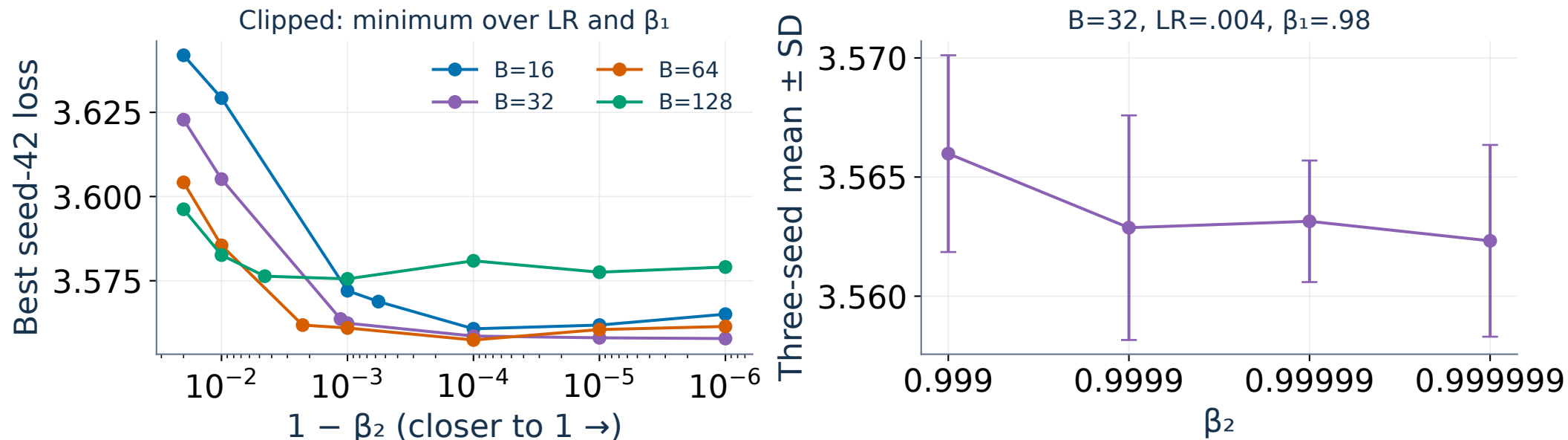
$$H_{\text{global}} = -\frac{B \cdot 512 \cdot \ln 2}{\ln \beta}$$



$\approx 0.56\text{M}$ global targets
Synchronous β_1 memory at $B = 16\text{--}64$

At fixed β , a smaller batch shortens memory in tokens. Increasing β can preserve the token horizon. This interpretation of selected recipes is not a validated universal scaling law.

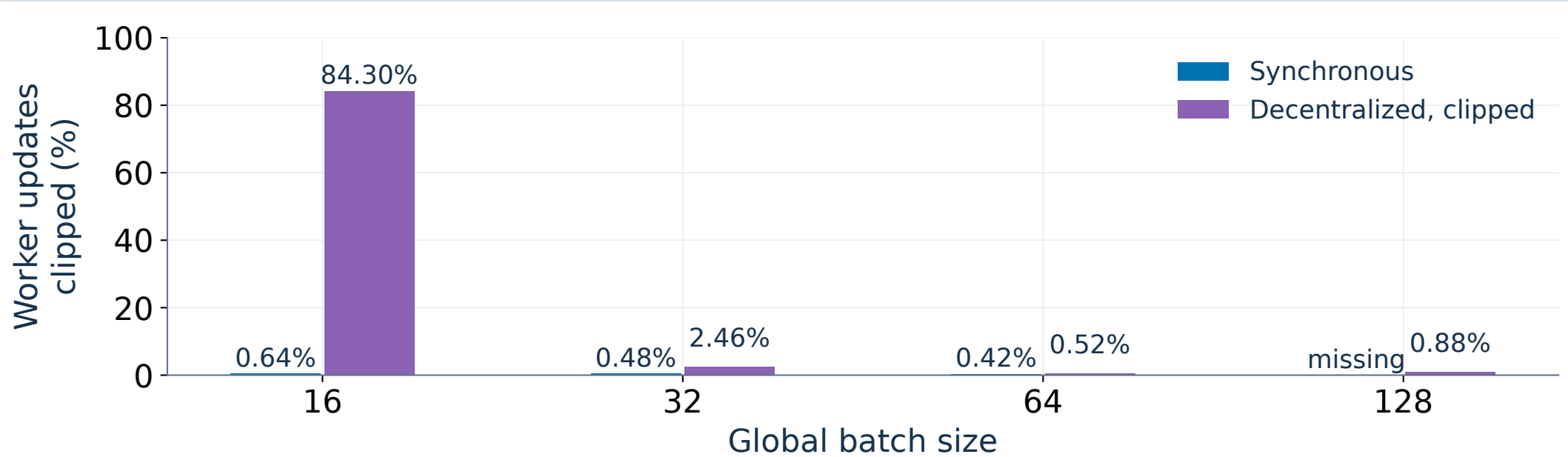
Long second-moment memory helps small batches



At batch 32, three β_2 choices near one differ by only 0.000819 in mean loss.

Profiles minimize over LR and β_1 using seed 42: screening diagnostics, not replicated estimates.
The β_2 trend is not monotonic; the largest tested value is not consistently best.

Small local batches clip frequently

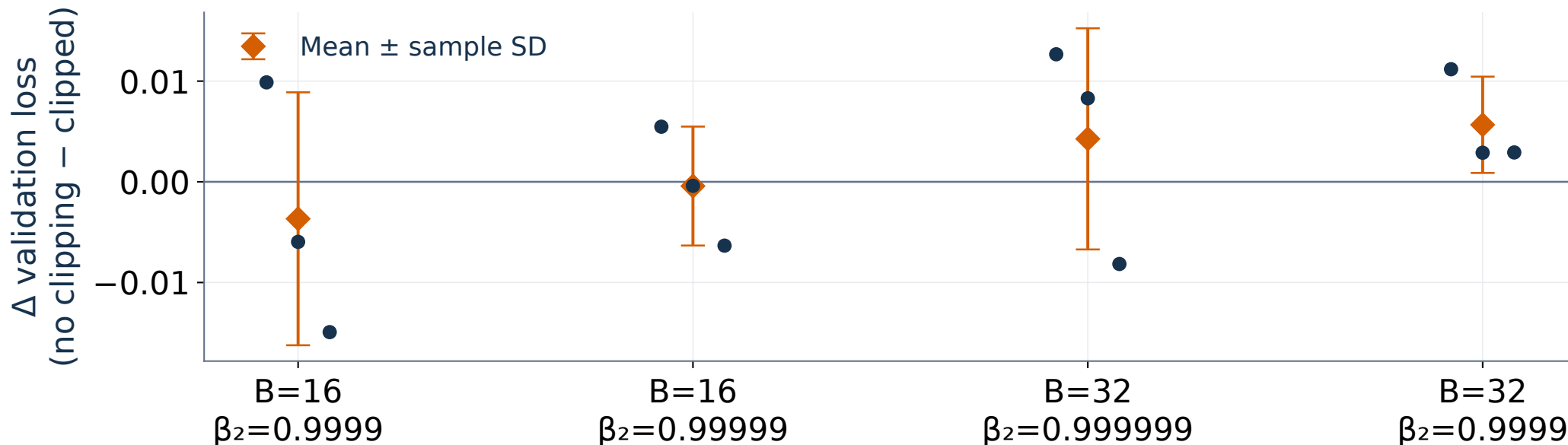


84.3% local batch 4

2.46% local batch 8

Threshold 1; mean over workers and seeds. Synchronous selected batches 16–64 clip on less than 0.64% of updates. Historical synchronous batch-128 counters are unavailable. Frequency does not establish benefit.

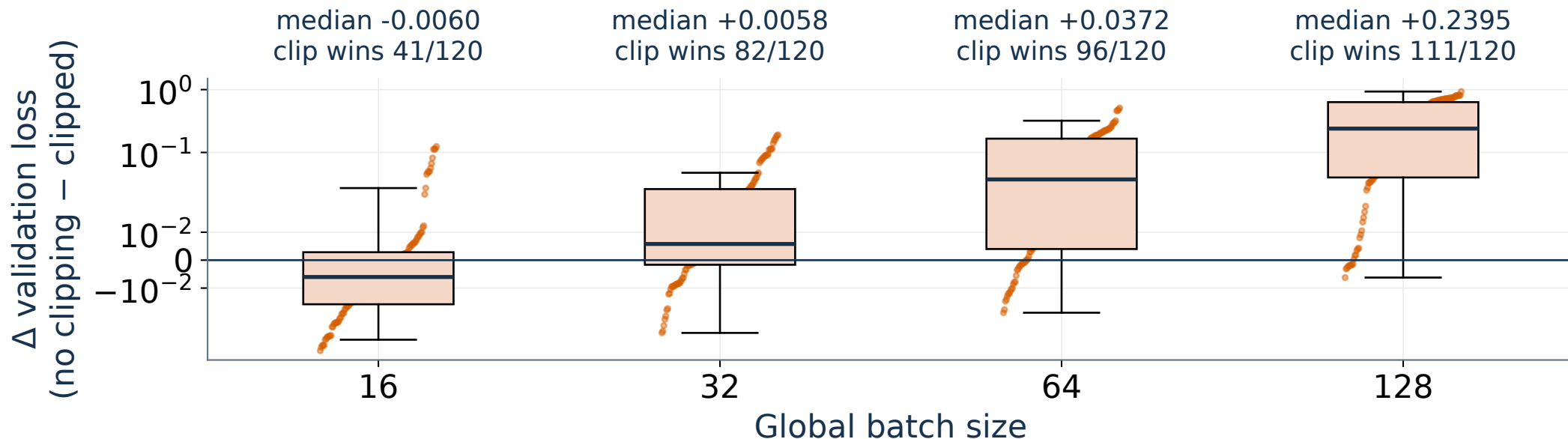
Frequent clipping need not be beneficial



Batch 16 can improve when clipping is removed; the original batch-32 recipes favor clipping.

Δ = no clipping – clipping; negative favors removal. All other recipe settings are held fixed. Four selected configurations $\times$ three seeds. Controls were selected on these seeds; this is not fresh confirmation.

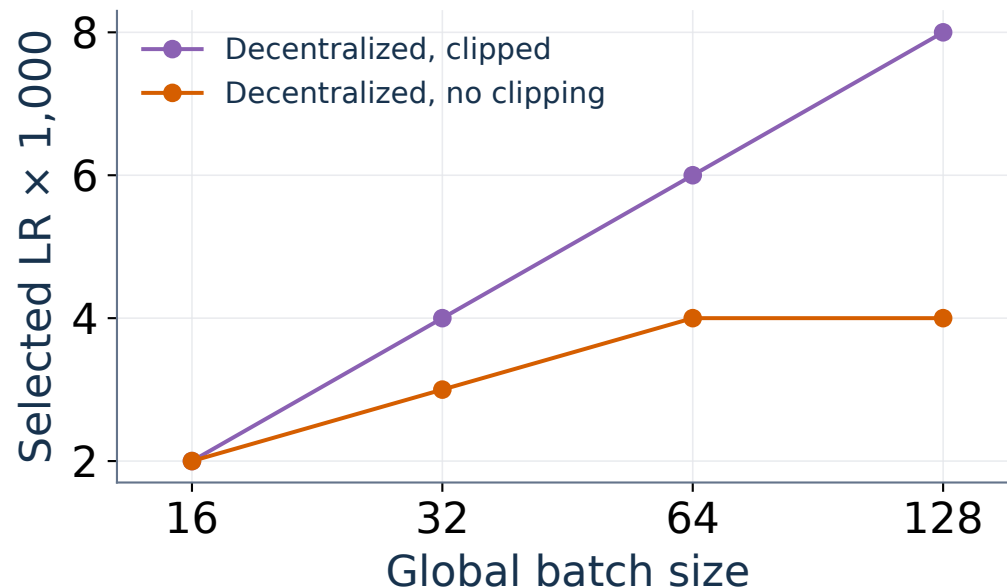
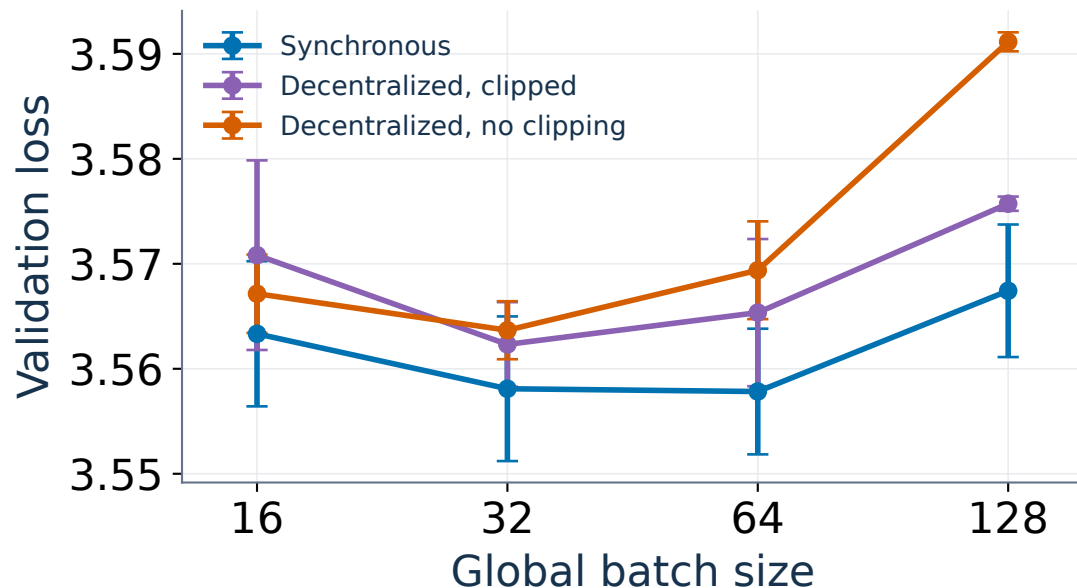
Clipping helps more across the larger-batch grids



No-clipping wins: 79/120 $\rightarrow$ 38/120 $\rightarrow$ 24/120 $\rightarrow$ 9/120 as batch increases.

All 480 matched seed-42 pairs succeeded. Symmetric-log loss axis; linear within ± 0.025 .
LR, β_1 , β_2 , batch, and seed match within each pair. Poor configurations skew mean differences.

Removing clipping changes the preferred LR

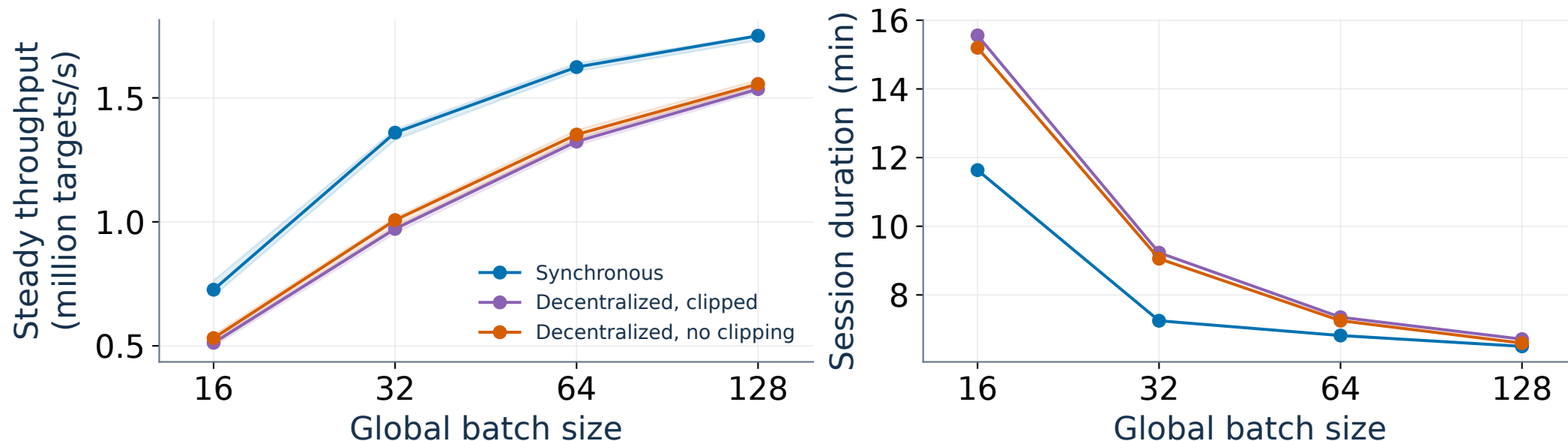


-0.003672 retuned Δ at batch 16

+0.015419 retuned Δ at batch 128

At batch 32, removal helps matched replicated LR=.003 recipes but hurts the original LR=.004 controls.
Clipping interacts with LR: separately retuned winners do not isolate its causal effect.

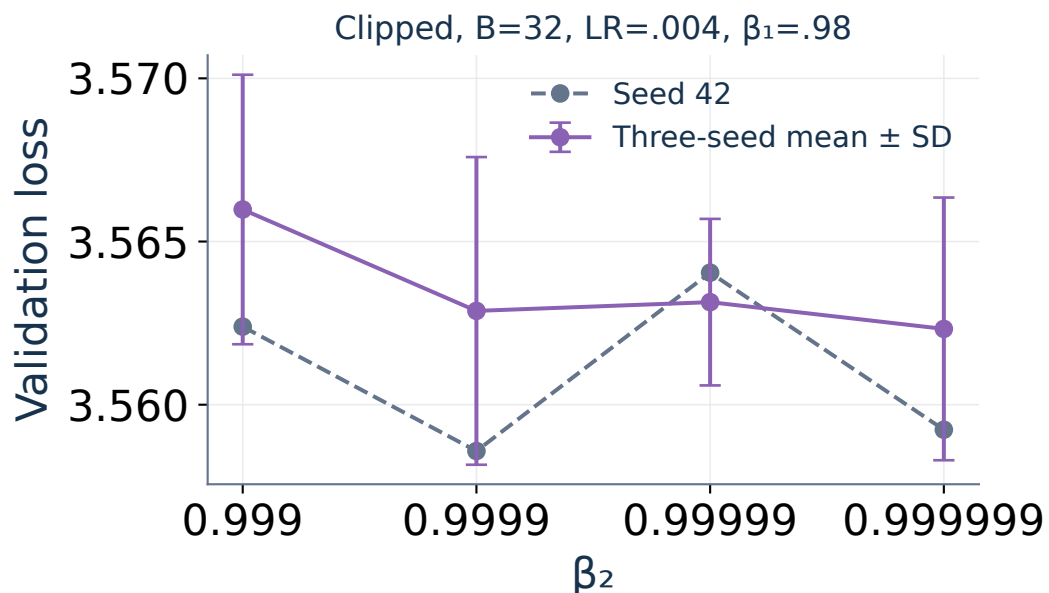
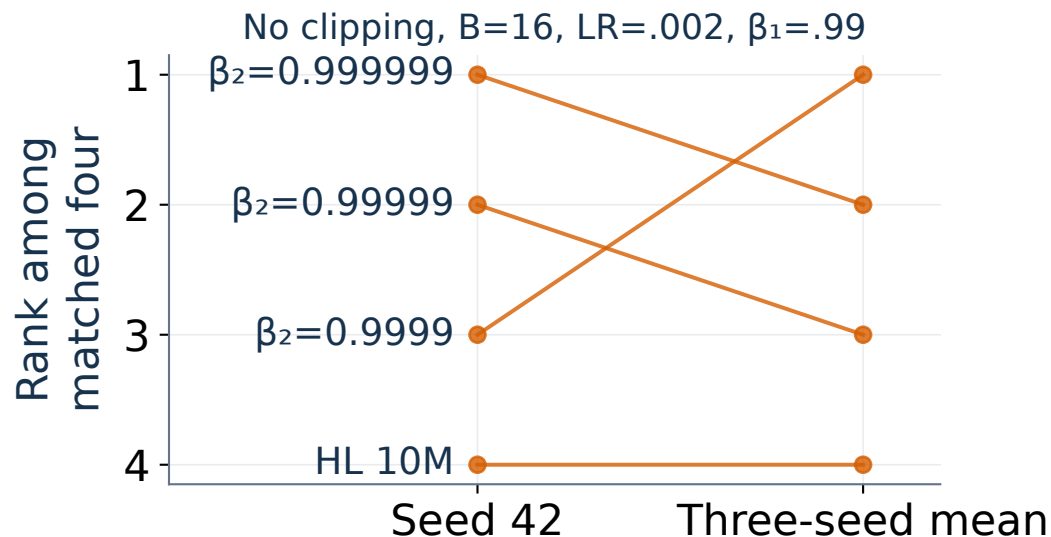
Better token efficiency has a throughput cost



A fixed training-token budget is not a fixed wall-clock budget.

Descriptive medians from the recorded execution stratum; configuration pools and job placements differ. Four logical workers use one GH200, not four physical GPUs. Memory peaks include batch-128 evaluation.

Replication changes fine-grained β_2 rankings



Batch 16 without clipping: $\beta_2=0.999999$ leads screening; $\beta_2=0.9999$ leads after replication.

Seeds 43–44 contribute to final ranking; they are not fresh confirmation. Remaining no-clipping endpoints: upper β_2 at batch 32; lower β_1 at 64 and 128. Neither endpoint establishes an optimum beyond the grid.

Five findings, one joint tuning problem

- 01 Small batches can improve quality.** The best observed batches are 32–64, not always 16.
- 02 Moment memory matters.** Smaller batches favor higher β_1 ; β_2 benefits plateau near one.
- 03 The decentralized gap is small.** Clipped recipes approach synchronous quality in this experiment.
- 04 Small local batches clip often.** This rate measures intervention frequency, not usefulness.
- 05 Clipping helps larger batches more here.** Its value still depends on LR and the tested grid.

Tune batch size, LR, β_1 , β_2 , and clipping together; judge quality and cost separately.

Lowest observed means: synchronous B64, 3.557833 · clipped decentralized B32, 3.562323.

Three-seed tuning results support these findings; they do not establish universal optima or statistical equivalence.